Born from causality, Tsirelson from steering, and the amplitude seam (SUBMISSION v3)

Born from causality, Tsirelson from steering, and the amplitude seam:

an internal reconstruction inside a geometric hidden-connection model

Cycle 3 draft v1. Author: A. Oktyabrev (aoktyabrev@gmail.com). Status: assembly draft (~11 pp + battery appendix). All numbers cite raw as `file (commit)`. Claim-discipline per program standard: theorem only with written proof + battery + review; conjecture otherwise.

Abstract

Within a geometric hidden-connection ("ribbon") model we close, and then delimit, an internal reconstruction of quantum correlations. (i) Where a mechanism carries **steering** — conditional states with internally generated diversity $D(\chi)>0$ — the no-signaling prohibition forces the Born exponent $p=2$ uniquely; the derivation is internal (the steering ensembles are produced by a shifted source measure, not imported), executable, and robust under five deformations of the measure. (ii) Mechanisms that resolve outcomes **locally** are capped by Bell's theorem at $S=2$, with a **1/3-cosine point** ($E=\cos\theta/3$) as their isotropic-cosine ceiling, anchored analytically and reached by two mechanisms, with ISO-DYN bracketing it. (iii) The **amplitude seam** between that ceiling and the quantum value $S=2\sqrt{2}$ is not crossed by decorating local mechanisms — three registered attempts fail on a trilemma of horns (no steering / no amplitude / superluminal telegraph) — but only by **joint, nonlocal resolution**, which by our foliation results demands a shared, statistically invisible precedence structure — a preferred foliation being one realization, a frameless shared order-coin another (class-M theorem). The bridge statement is an exact division of labor, not a unification.

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1. Introduction — the import and the question

The program's canonical text acknowledged an **import**: partially entangled ensembles were introduced by hand ("within this ansatz… we make no claim of a general derivation of the Born rule", FINAL_v1:284–290). Cycle 3 asks whether the steering that underlies the Born rule can be **generated internally** by the model's own geometry, and — once it is — what exactly stands between such a mechanism and full quantum correlations. The answer assembles into four registered statements (RULE, CEILING, SEAM, FRAME), each with kill-criteria and public commits.

2. Method frame

This work is preregistered. Each campaign registers predictions, estimators, and **kill criteria** (owned by the operator) *before* raw data; the pipeline is raw → commit → analysis, with executable prereg↔JSON assert checklists run before every battery. Batteries are pure numpy ($N=2\cdot10^6$ unless noted), PRNGKey from config; analytics sit beside numerics (a $>2\sigma$ mismatch is a stop). Claim-discipline: **theorem** only with a written proof, a battery, and operator review; **conjecture** otherwise.

Reproduction. The full numpy battery suite of cycle 3 reproduces bitwise: re-executed with fixed seeds, every results JSON is byte-identical under `git diff` (C3-B B1–B7, hardening, L3, L1, L2, C3-S, β-coin), and C3-B and the β-coin were independently reproduced by the operator in a separate environment.

Self-correction register (a feature, not dirty laundry). Every false stop and retraction below was caught by a preregistered criterion or a literature check, not by hindsight; all are public with commits.

#	What	Caught by	Resolution	Commit
1	$\Delta(2)>0$ (normalized conditional axis broke total probability)	“$\Delta(2)>0$ = bug” criterion	raw sub-unit Bloch vector	968aaa5
2	Quadrature-tolerance gate (absent from prereg) false-stopped T1/T2	prereg↔registered criterion	$<2\sigma$ criterion + algebraic zero at $p=2$; no threshold loosened	ccf5af6
3	Monte-Carlo cross-range would false-stop the foliation test	control design	common-random-number comparison	49e8c1b
4	§2.3 over-claim (“no-signaling alone bounds Tsirelson”)	battery rediscovered the PR-box fact [PR94]	narrowed to steering-endowed layers	2053106

#	What	Caught by	Resolution	Commit
5	Frame over-claim (“preferred frame required”)	class-M theorem T3′ + β-coin	“shared statistically invisible precedence structure”; foliation one realization	c8e1bf3

Section 8 gives the detailed methods and controls; this frame is the summary a reader needs first.

3. Model

Source / steering (F2). A shared connection orientation λ on S^2 is drawn from a shifted measure $\mu_\chi \propto 1+\chi\lambda_z$. A measurement setting a partitions λ by $s=\mathrm{sign}(\lambda\cdot a)$; the conditional ensemble’s **raw** (sub-unit) mean vector m_s is the steered state; readout probability f_p(c)=|1+c|^{p/2}/(|1+c|^{p/2}+|1−c|^{p/2}), with f_∞ the step 1{c>0}. **Lesson (record):** normalizing the conditional axis broke total probability and produced Δ(p=2)>0 — the “Δ(2)>0 = bug” criterion caught it; the raw sub-unit Bloch vector is correct (battery_c3b, 968aaa5).

Joint / chord layer. A postulated pair law $P(s,t|a,b)=(1-st\cos\theta)/4$ ($E=-\cos\theta$, $S_{max}=2\sqrt{2}$), and its weighted generalization $|s\sqrt{w_a}\, a - t\sqrt{w_b}\, b|^p$, used where the mechanism resolves the pair jointly rather than by local readout of λ.

4. RULE — Born from no-signaling, internally (C3-B)

Steering is generated, not imported. With μ_χ, χ>0 yields steering diversity D(χ): 0.03→0.08 as χ grows (C3B_hardening.json, 968aaa5) — remote ensembles with **equal means, distinct higher moments**.

Selection lemma (executable). For the readout as a random variable over the conditioning, no-signaling ⇔ the readout is affine in the projection (Jensen gap of an affine map ≡ 0). Affinity ∩ oddness ($f(-c)=1-f(c)$) ∩ normalization = the single rule $f_2=(1+c)/2$ ⇒ **p=2 uniquely**. The algebraic zero at p=2 is exact (addendum2, ccf5af6); the non-affine tail is strictly positive — scan Δ(4)=0.042, Δ(6)=0.105, Δ(10)=0.217, Δ(∞)=0.375, monotone, analytic↔numeric <2σ (C3L_L3.json, 9567ddb).

Hardened to theorem. T1 (sequential/order, B7): repeatability 1.0000, order commute, no-signaling both orders, post-measurement null-set {2}. T2 (anti-circularity): **five** measure deformations (μ_κ, κ∈{0.5,2,3}; two cubic), all D>0, analytic null-set {2}, |analytic−numeric|<2σ. T3 (F1 direct law) is a boundary (postulated, not steering-collapse). Verdict: **internal derivation of the Born rule from no-signaling within the ribbon’s steering class** (C3B_hardening.json, c152c98). One **false stop** en route — a quadrature-tolerance gate absent from preregistration — was caught, documented, and corrected without loosening any registered criterion (addendum2, ccf5af6; §8).

Mechanical realization (C3-B-mech, form-free). The steering premise is realized mechanically and form-free: an annealed, field-magnetized and fast-frozen elastic source supplies a polar-biased orientation measure that fits no registered closed form; no closed form is needed. The sign-readout conditionals of that measure carry steering diversity in the third and fifth moments, matching the Jensen analytics computed directly on the empirical ensembles, with the unbiased ($h=0$) control showing null effects throughout; on ensembles produced entirely by the elastic dynamics, no-signaling selects $p=2$. The selection theorem's premise thus rests on measured moments, not on an assumed measure — a weaker assumption and hence a stronger claim (C3Bmech_M1M2.json, 2080bbe; M1✓M2✓ in form-free mode). That the source measure admits no closed form is kin to the C2-TM wall (the relaxational amplitude A(k_f) likewise has no closed form): an observation — elasticity does not favor closed forms — not a theorem.

5. CEILING — the 1/3-cosine point

Local resolution (hidden λ + local linear readout) yields $E(a,b)=E\_\lambda[(\lambda\cdot a)(\lambda\cdot b)]=a^T(⅓I)b$ = $\mathbf{\cos\theta/3}$ (χ-corrections odd ⇒ vanish), $S=2\sqrt{2}/3\approx0.943$ — the isotropic-cosine ceiling of the local layer. The exact convergence is analytic (lever rule, exactly 1/3) and frame-local ($E(0)=0.333$; C3L_L2.json, 49e8c1b); dynamical isotropization lands NEAR the point: $\rho\_cos=0.311\pm0.010$ (cosine-fit, the form-matched estimator; raw C2ISO_analysis.json: cells/kf4.0/rho_cos = 0.3113, a043f8f), ρ_model-free=0.336 (same file) — bracketing 1/3 within ~2σ. Estimators named per program rule. Anchored analytically and reached by two mechanisms, with ISO-DYN bracketing it ⇒ a structural boundary, the lower bank of the seam.

6. SEAM — the trilemma of horns (C3-S)

Can an internal model reach **D>0 and S=2√2** together? Registered attempts, ansätze fixed before batteries: - **S-F1** (weighted chord, $\alpha=\sqrt{(1+\chi)}$, $\beta=\sqrt{(1-\chi)}$): $S(\chi)=2\sqrt{2}\cdot\sqrt{(1-\chi^2)}$ — reaches $2\sqrt{2}$ only at $\chi=0$ (symmetry), but is postulated (no continuous measure) ⇒ D_joint N/A. **Horn: no internal steering.** - **S-F2** (shifted source × chord/product mixture): $S(\chi)=\sqrt{2}\cdot(1+\chi/3)\leq1.886$, and $P(s|a,\lambda)=½$ ⇒ Alice does not update λ ⇒ D_joint≈0. **Horn: no amplitude / no steering** (C3S.json, 324f097). - **S-F3** (local bias × chord kernel, |·| ansatz with κ): steers (D>0) but the marginal carries Bob-setting dependence at $O(\kappa\chi)$ — telegraph 0.500→0.542 (C3S_F3_check.py; analytics 2053106). **Horn: superluminal telegraph** — kin to finite-speed no-gos [BPA12].

Reduction (Bell corollary, not a new theorem). The internal class = λ-average of local responses = LHV ⇒ $S\leq2$. Closing the seam therefore demands **joint, nonlocal resolution**; and joint resolution demands a shared, statistically invisible precedence structure (§7). Attempt count: 3. (C3S_seam_reduction.md.)

7. FRAME — foliation (C3-L, L1/L2)

On the chord joint law, two resolution mechanics: - **(i) preferred-foliation:** $S=2\sqrt{2}$ **identical across five foliations** (cross-disc 0.000000 under common random numbers), foliation statistically invisible (C3L_L2.json, 49e8c1b). - **(ii) frame-local:** space-like → LHV ceiling $S=0.943$; time-like A→B, B→A → reconstruct $2\sqrt{2}$. L1: the joint admits an order-invariant realization (existential PASS, factorized witness disc=0.00); the collapse realization's order-dependence (0.0418 with invariant marginals) is evidence for L2c, not pathology (addendum1).

Method note: the foliation-invariance test required common random numbers — a raw cross-range would false-STOP on Monte-Carlo noise (§8).

Class-M theorem (status: theorem; canonical statement in `sim/cycle3/C3L_L2c_THEOREM.md`, c8e1bf3): in the class M of outcome-definite sequential resolution mechanisms (shared preparation randomness λ, setting-independence M4′, no retrocausation), S>2 is achievable **iff** M admits a shared, space-like-transcending, statistically invisible precedence structure. Realizations demonstrated: a preferred foliation (five axes, cross-discrimination 0.000000 under CRN) and a **frameless per-run shared order-coin** (S=2√2±2σ, no telegraph, coin invisible; battery c8e1bf3, independently re-executed). The earlier conjecture's "preferred foliation is required" was **retracted as too strong** during the adversarial passes: the foliation is one realization of the required structure, not its only form. Experimental anchor unchanged: before-before experiments [ZBGT01, SZGS02] exclude the frame-local branch in nature; what survives is joint resolution carrying a shared, hidden precedence structure. Kin: Bohmian preferred slicing [DGZ92], steering-axiomatic reconstructions [CDP11]; [H92], [SS97].

8. METHODS — batteries, controls, self-correction (detail; summary in §2)

All batteries: pure numpy, $N=2\cdot 10^6$, PRNGKey from config, analytics beside numerics (>2σ mismatch = stop), assert prereg↔JSON before raw, raw→commit→analysis. Estimator named in every prediction; metric = the model observable.

Registered false stops and their resolution (part of the evidence): 1. **Δ(2)>0 = bug** (C3-B): normalized conditional axis broke total probability; fixed to raw Bloch (968aaa5). 2. **Quadrature gate** (hardening): an analytic-tolerance gate (tol=1e-9 on a 20001-pt trapezoid, error ~2.3e-5) false-STOPped T1/T2; corrected to the registered <2σ criterion + algebraic zero at p=2; threshold **not** loosened to pass (addendum2, ccf5af6). 3. **Monte-Carlo cross-range** (L2): raw 5-foliation range would false-STOP; replaced by common-random-number comparison (49e8c1b). 4. **§2.3 over-claim retraction** (C3-S): the first §2.3 asserted "no-signaling alone cuts p>2 in the joint layer"; J5 rediscovered the Popescu–Rohrlich fact that the postulated |·|^p family is no-signaling up to the PR box [PR94]; the claim was narrowed to steering-endowed layers (324f097; §9). 5. **Frame over-claim** (abstract/§5/§8/§9 v1): "preferred frame required" — retracted by the class-M theorem's T3′ and the β-coin demonstration; revised in v2, history retained. §9 CEILING wording aligned in v2. Each was caught by a preregistered criterion, not by hindsight.

Reproduction. The full numpy battery suite of cycle 3 was independently re-executed (2026-07-21); fixed-seed raw reproduces bitwise (C3-B B1–B7, C3-B hardening, L3, L1, L2, C3-S, β-coin — all results JSON byte-identical under `git diff`). C3-B and the β-coin were additionally reproduced by the architect in a separate environment.

9. The Tsirelson section (§2.3, corrected — verbatim)

> Rule selection is exactly as strong as steering. In any layer whose measurements produce conditional states that can themselves be steered — internally generated diversity D>0 — the Jensen argument applies: no-signaling forces affinity, selects p=2, and with it the quantum value of S

available to that layer. In a bare-correlation joint layer, however, where outcomes carry no post-measurement structure, the $|\cdot|^p$ family has uniform marginals for every p: the entire family is no-signaling, up to and including the PR box — the classic observation of Popescu and Rohrlich [PR94], which our battery rediscovered as a registered finding against an earlier over-claim of this section (retained in the record). No-signaling alone therefore does not enforce the Tsirelson bound; what enforces it, in this geometry, is no-signaling PLUS steering structure — and this localizes, within one construction, the same gap that information causality [Paw09] was invented to fill. The amplitude seam then resolves into a reduction rather than a mystery: within the internally-generated class (hidden measure plus local response) Bell's theorem itself caps S at 2, so closing the seam — D>0 together with $S=2\sqrt{2}$ — demands joint, nonlocal resolution of the pair; and joint resolution, by the foliation results above, demands a preferred frame that the statistics provably hide. [Revised by the class-M theorem, §7: read 'a shared, statistically invisible precedence structure', of which a preferred frame is one realization — the frameless order-coin is another.] The program's bridge statement follows: Einstein's prohibition selects the quantum rule wherever mechanisms carry steering; the quantum amplitude is bought only by joint resolution, whose frame Einstein's own statistics render invisible.

10. Outlook (verbatim)

Three cycles of this program asked one question from three sides: what stands between a mechanism and quantum correlations. The answer assembled itself into four registered statements. (1) RULE: wherever a mechanism carries steering — conditional states with internally generated diversity — the prohibition of superluminal signaling forces the Born exponent p=2 uniquely; the derivation is internal, executable, and robust to deformations of the source measure. (2) CEILING: mechanisms that resolve outcomes locally (hidden measure plus local response) are capped by Bell's theorem at S=2, with the 1/3-cosine point as their isotropic-cosine ceiling — a landmark anchored analytically (the lever rule), reached by frame-local resolution, and bracketed by relaxational dynamics (estimator-dependent, §5). (3) SEAM: the gap between that ceiling and the quantum value is not crossed by decorating local mechanisms — our registered attempts fail on a trilemma of horns (no steering, no amplitude, or superluminal telegraph; kin to finite-speed no-gos [BPA12]) — it is crossed only by joint resolution of the pair. (4) FRAME: joint resolution of the frame-local kind reconstructs the Suarez–Scarani alternative and degrades to the local ceiling, which real before-before experiments [ZBGT01, SZGS02] have excluded in nature; the surviving corner, joint resolution carrying a shared, statistically invisible precedence structure (foliation or shared order-coin), reproduces quantum statistics exactly while rendering its own precedence structure statistically invisible to the resolution of our batteries. The bridge statement of the program is therefore not a unification but an exact division of labor: Einstein's causality selects the quantum rule; the quantum amplitude is bought only by joint resolution; and the precedence structure that joint resolution requires is precisely what the selected statistics forever hide.

What remains open is stated as precisely: a mechanical realization of the asymmetry parameter (C3-B-mech), the structural theorem behind the seam trilemma (the class-M theorem, proved; N-event generalization = C4 G-T), and the ground-state problem of the stiff chain — walls mapped, kill criteria attached, commits public.

A fourth cycle extended the bridge to N parties. A value theorem at N=3: Mermin $M_3 > 2$ is achievable in class M iff at least one space-like pair is ordered — a single ordered pair suffices for the algebraic maximum. The tripartite mirror of the bipartite picture is complete: internal geometry (a shared parity invariant — the ribbon's Z_2 holonomy) yields the GHZ signature at the Mermin settings while capped at the classical bound (the tripartite seam, registered at prior 0.45 before the battery), and one ordered pair crosses it, reproducing the complete GHZ signature at those settings. The currency of genuine N-partite nonlocality is binary in block size: disjoint ordered pairs cap the Mermin-Klyshko value at the 2-producible ceiling for N ≥ 4 and at the Svetlichny bound, while only a full precedence block attains the top tier — algebraic (4.0 at N = 4), above the quantum 2^{(N−1)/2}. What cuts class M from the algebraic to the quantum tier at N parties — the bipartite answer was no-signaling plus steering structure — is the program's registered next open problem. Schedule-invisibility within a precedence structure stands as a theorem: the achieved tier certifies the structure; no statistic reveals the schedule. This cycle's registered retractions (a k-pair "climbing ladder", an over-scoped GHZ claim, a "45°-tail" construction) are part of the public record, per the method frame of §2.

Appendix A — batteries (reproducible)

Battery	Prereg	Raw	Result
C3-B B1–B7	dd7a74e	968aaa5	steering generated, p=2 forced
C3-B hardening T1–T3	96a3178	c152c98	internal derivation (T1✓T2✓)
L3 scan + bridge	2d78c51	9567ddb	tail Δ>0; internal S(2)=√2
C3-L L1/L2	0b10f5d,100a261	1e0cf0a,49e8c1b	foliation invisible; frame-local→ceiling
C3-S J1–J7	3d516d4	324f097	seam not closed (S-F1/S-F2)
S-F3 analytics	—	2053106	closed pre-battery (telegraph)
β-coin (class M)	C3L_coin_prereg	c8e1bf3	frameless 2√2, invisible
Class-M theorem	—	c8e1bf3	achievability iff shared invisible precedence
C3-B-mech M0 (calibration)	C3Bmech addenda	0a6c51e	source polarizes; χ(h) linear

Battery	Prereg	Raw	Result
C3-B-mech M1/M2	63456cd	2080bbe	mechanical steering, form-free (M1✓M2✓)

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